1) a)

$\operatorname{cosech} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}} = \frac{2e^x}{e^{2x}-1}$

1) b)

$\int \operatorname{cosech} x \, dx = \int \frac{2e^x}{e^{2x}-1} dx$

Use Integration by Substitution:

Let $u = e^x \Rightarrow \frac{du}{dx} = e^x \Rightarrow dx = \frac{1}{e^x} du$

$\therefore \int \frac{2e^x}{e^{2x}-1} dx = \int \frac{2}{u^2-1} du = -2\int \frac{1}{1-u^2} du$

$= -2 \operatorname{artanh} u + c$

$= -2 \operatorname{artanh} e^x + c$

2) a)

$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}} = \frac{2e^x}{e^{2x}+1}$

2) b)

$\int \operatorname{sech} x \, dx = \int \frac{2e^x}{e^{2x}+1} dx$

Use Integration by Substitution:

Let $u = e^x \Rightarrow \frac{du}{dx} = e^x \Rightarrow dx = \frac{1}{e^x} du$

$\therefore \int \frac{2e^x}{e^{2x}+1} dx = \int \frac{2}{u^2+1} du$

$= 2 \arctan u + c$

$= 2 \arctan e^x + c$

3)

First find where the two curves intersect.

$12 \operatorname{sech} x = 24 - 9 \cosh x$

$\Rightarrow \frac{4}{\cosh x} = 8 - 3 \cosh x$

$\Rightarrow 4 = 8 \cosh x - 3 \cosh^2 x$

$\Rightarrow 3 \cosh^2 x - 8 \cosh x + 4 = 0$

$\Rightarrow (3 \cosh x - 2)(\cosh x - 2) = 0$

$\Rightarrow \cosh x = \frac{2}{3}$ or $\cosh x = 2$

$\cosh x \neq \frac{2}{3}$ as $\cosh x \geq 1$ for all real values of x

So $\cosh x = 2 \Rightarrow x = \pm\ln\left(2+\sqrt{2^2-1}\right) = \pm\ln\left(2+\sqrt{3}\right)$

$-\ln\left(2+\sqrt{3}\right) = \ln\left(2+\sqrt{3}\right)^{-1} = \ln\left(2-\sqrt{3}\right)$

Now integrate:

$I = \int_{\ln(2-\sqrt{3})}^{\ln(2+\sqrt{3})}(24-9\cosh x)\,\mathrm{d}x - \int_{\ln(2-\sqrt{3})}^{\ln(2+\sqrt{3})} 12\operatorname{sech} x\,\mathrm{d}x$

$= \int_{\ln(2-\sqrt{3})}^{\ln(2+\sqrt{3})}(24-9\cosh x-12\operatorname{sech} x)\,\mathrm{d}x$

Note: $\int \operatorname{sech} x\,\mathrm{d}x = \int \frac{2\mathrm{e}^x}{\mathrm{e}^{2x}+1}\mathrm{d}x$

Use Integration by Substitution:

Let $u = \mathrm{e}^x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \mathrm{e}^x \Rightarrow \mathrm{d}x = \frac{1}{\mathrm{e}^x}\mathrm{d}u$

$\therefore \int \frac{2\mathrm{e}^x}{\mathrm{e}^{2x}+1}\mathrm{d}x = \int \frac{2}{u^2+1}\mathrm{d}u$

$= 2\arctan u + c$

$= 2\arctan \mathrm{e}^x + c$

$\therefore I = [24x - 9\sinh x - 24\arctan \mathrm{e}^x]_{\ln(2-\sqrt{3})}^{\ln(2+\sqrt{3})}$

$= \left(24\ln\left(2+\sqrt{3}\right) - 9\sinh(\ln\left(2+\sqrt{3}\right) - 24\arctan \mathrm{e}^{\ln(2+\sqrt{3})}\right) - \left(24\ln\left(2-\sqrt{3}\right) - 9\sinh(\ln\left(2-\sqrt{3}\right) - 24\arctan \mathrm{e}^{\ln(2-\sqrt{3})}\right)$

$= \left(24\ln\left(2+\sqrt{3}\right) - 9\times\frac{\mathrm{e}^{\ln(2+\sqrt{3})}-\mathrm{e}^{-\ln(2+\sqrt{3})}}{2} - 24\arctan\left(2+\sqrt{3}\right)\right) - \left(24\ln\left(2-\sqrt{3}\right) - 9\times\frac{\mathrm{e}^{\ln(2-\sqrt{3})}-\mathrm{e}^{-\ln(2-\sqrt{3})}}{2} - 24\arctan\left(2-\sqrt{3}\right)\right)$

$= \left(24\ln\left(2+\sqrt{3}\right) - \frac{9}{2}\left(\mathrm{e}^{\ln(2+\sqrt{3})} - \mathrm{e}^{\ln(2+\sqrt{3})^{-1}}\right) - 24\times\frac{5\pi}{12}\right) - \left(24\ln\left(2-\sqrt{3}\right) - \frac{9}{2}\left(\mathrm{e}^{\ln(2-\sqrt{3})} - \mathrm{e}^{\ln(2-\sqrt{3})^{-1}}\right) - 24\times\frac{\pi}{12}\right)$

$= \left(24\ln\left(2+\sqrt{3}\right) - \frac{9}{2}\left(\mathrm{e}^{\ln(2+\sqrt{3})} - \mathrm{e}^{\ln(2-\sqrt{3})}\right) - 10\pi\right) - \left(24\ln\left(2-\sqrt{3}\right) - \frac{9}{2}\left(\mathrm{e}^{\ln(2-\sqrt{3})} - \mathrm{e}^{\ln(2+\sqrt{3})}\right) - 2\pi\right)$

$= \left(24\ln\left(2+\sqrt{3}\right) - \frac{9}{2}\left(2+\sqrt{3}-\left(2-\sqrt{3}\right)\right) - 20\pi\right) - \left(24\ln\left(2-\sqrt{3}\right) - \frac{9}{2}\left(2-\sqrt{3}-\left(2+\sqrt{3}\right)\right) - 2\pi\right)$

$= \left(24\ln\left(2+\sqrt{3}\right) - 9\sqrt{3} - 10\pi\right) - \left(24\ln\left(2-\sqrt{3}\right) + 9\sqrt{3} - 2\pi\right)$

$= 24\ln\left(\frac{2+\sqrt{3}}{2-\sqrt{3}}\right) - 18\sqrt{3} - 8\pi$

$= 24\ln\left(7+4\sqrt{3}\right) - 18\sqrt{3} - 8\pi$

4) a)

$$\int \tanh x \, \mathrm{d}x = \int \frac{\sinh x}{\cosh x} \mathrm{d}x = \ln|\cosh x| + c$$

4) b)

$$\int \coth x \, \mathrm{d}x = \int \frac{\cosh x}{\sinh x} \mathrm{d}x = \ln|\sinh x| + c$$

5) a)

$$\int \operatorname{cosech} x \coth x \, \mathrm{d}x$$
$$= \int \frac{1}{\sinh x} \times \frac{\cosh x}{\sinh x} \mathrm{d}x$$
$$= \int \cosh x \, (\sinh x)^{-2} \, \mathrm{d}x$$
$$= -(\sinh x)^{-1} + c$$
$$= -\operatorname{cosech} x + c$$

Alternatively:

$$\int \operatorname{cosech} x \coth x \, \mathrm{d}x$$
$$= \int \frac{2}{\mathrm{e}^x - \mathrm{e}^{-x}} \times \frac{\mathrm{e}^x + \mathrm{e}^{-x}}{\mathrm{e}^x - \mathrm{e}^{-x}} \mathrm{d}x$$
$$= \int \frac{2\mathrm{e}^x + 2\mathrm{e}^{-x}}{(\mathrm{e}^x - \mathrm{e}^{-x})^2} \mathrm{d}x$$

Use Integration by Substitution:

Let $u = \mathrm{e}^x - \mathrm{e}^{-x} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \mathrm{e}^x + \mathrm{e}^{-x} \Rightarrow \mathrm{d}x = \frac{1}{\mathrm{e}^x + \mathrm{e}^{-x}} \mathrm{d}u$

$$\int \frac{2\mathrm{e}^x + 2\mathrm{e}^{-x}}{(\mathrm{e}^x - \mathrm{e}^{-x})^2} \mathrm{d}x = \int \frac{2}{u^2} \mathrm{d}u = \int 2u^{-2} \, \mathrm{d}u = -2u^{-1} + c$$
$$= -\frac{2}{\mathrm{e}^x - \mathrm{e}^{-x}} + c = -\frac{1}{\sinh x} + c = -\operatorname{cosech} x + c$$

5) b)

$\int \operatorname{sech} x \tanh x \, \mathrm{d}x$

$= \int \frac{1}{\cosh x} \times \frac{\sinh x}{\cosh x} \mathrm{d}x$

$= \int \sinh x \, (\cosh x)^{-2} \, \mathrm{d}x$

$= -(\cosh x)^{-1} + c$

$= -\operatorname{sech} x + c$

Alternatively:

$\int \operatorname{sech} x \tanh x \, \mathrm{d}x$

$= \int \frac{2}{\mathrm{e}^x + \mathrm{e}^{-x}} \times \frac{\mathrm{e}^x - \mathrm{e}^{-x}}{\mathrm{e}^x + \mathrm{e}^{-x}} \mathrm{d}x$

$= \int \frac{2\mathrm{e}^x - 2\mathrm{e}^{-x}}{(\mathrm{e}^x + \mathrm{e}^{-x})^2} \mathrm{d}x$

Use Integration by Substitution:

Let $u = \mathrm{e}^x + \mathrm{e}^{-x} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \mathrm{e}^x - \mathrm{e}^{-x} \Rightarrow \mathrm{d}x = \frac{1}{\mathrm{e}^x - \mathrm{e}^{-x}} \mathrm{d}u$

$\int \frac{2\mathrm{e}^x - 2\mathrm{e}^{-x}}{(\mathrm{e}^x + \mathrm{e}^{-x})^2} \mathrm{d}x = \int \frac{2}{u^2} \mathrm{d}u = \int 2u^{-2} \, \mathrm{d}u = -2u^{-1} + c$

$= -\frac{2}{\mathrm{e}^x + \mathrm{e}^{-x}} + c = -\frac{1}{\cosh x} + c = -\operatorname{sech} x + c$

6) a)

$\cosh^2 x - \sinh^2 x = \left(\frac{\mathrm{e}^x + \mathrm{e}^{-x}}{2}\right)^2 - \left(\frac{\mathrm{e}^x - \mathrm{e}^{-x}}{2}\right)^2$

$= \left(\frac{\mathrm{e}^{2x} + 2 + \mathrm{e}^{-2x}}{4}\right) - \left(\frac{\mathrm{e}^{2x} - 2 + \mathrm{e}^{-2x}}{4}\right)$

$= \frac{\mathrm{e}^{2x} + 2 + \mathrm{e}^{-2x} - \mathrm{e}^{2x} + 2 - \mathrm{e}^{-2x}}{4}$

$= \frac{4}{4} = 1$ as req.

6) b)

$\int \operatorname{cosech}^2 x \, \mathrm{d}x$

$= \int \frac{1}{\sinh^2 x} \mathrm{d}x$

$= \int \frac{\cosh^2 x - \sinh^2 x}{\sinh^2 x} \mathrm{d}x$

$= -\int \frac{\sinh^2 x - \cosh^2 x}{\sinh^2 x} \mathrm{d}x$

$= -\int \frac{\sinh x \times \frac{\mathrm{d}}{\mathrm{d}x}(\cosh x) - \cosh x \times \frac{\mathrm{d}}{\mathrm{d}x}(\sinh x)}{(\sinh x)^2} \mathrm{d}x$

$= -\frac{\cosh x}{\sinh x} + c$

$= -\coth x + c$

Alternatively:

$\int \operatorname{cosech}^2 x \, \mathrm{d}x$

$= \int \left(\frac{2}{\mathrm{e}^x - \mathrm{e}^{-x}}\right)^2 \mathrm{d}x$

$= \int \frac{4}{(\mathrm{e}^x - \mathrm{e}^{-x})^2} \mathrm{d}x$

Use Integration by Substitution:

Let $u = \mathrm{e}^x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \mathrm{e}^x \Rightarrow \mathrm{d}x = \frac{1}{\mathrm{e}^x}\mathrm{d}u = \frac{1}{u}\mathrm{d}u$

$\int \frac{4}{(\mathrm{e}^x - \mathrm{e}^{-x})^2} \mathrm{d}x = \int \frac{4}{\left(u - \frac{1}{u}\right)^2} \times \frac{1}{u} \mathrm{d}u = \int \frac{4}{u^2 - 2 + \frac{1}{u^2}} \times \frac{1}{u} \mathrm{d}u = \int \frac{4u^2}{u^4 - 2u^2 + 1} \times \frac{1}{u} \mathrm{d}u = \int \frac{4u}{(u^2 - 1)^2} \mathrm{d}u$

Use Integration by Substitution:

Let $v = u^2 - 1 \Rightarrow \frac{\mathrm{d}v}{\mathrm{d}u} = 2u \Rightarrow \mathrm{d}u = \frac{1}{2u}\mathrm{d}v$

$\int \frac{4u}{(u^2 - 1)^2} \mathrm{d}u = \int \frac{2}{v^2} \mathrm{d}u = \int 2v^{-2} \, \mathrm{d}u = -2v^{-1} + c$

$= -\frac{2}{u^2 - 1} + c$

$= -\frac{2}{\mathrm{e}^{2x} - 1} + c$

$= -\frac{2}{\mathrm{e}^{2x} - 1} - 1 + C$

$= -\frac{2}{\mathrm{e}^{2x} - 1} - \frac{\mathrm{e}^{2x} - 1}{\mathrm{e}^{2x} - 1} + C$

$= -\left(\frac{2}{\mathrm{e}^{2x} - 1} + \frac{\mathrm{e}^{2x} - 1}{\mathrm{e}^{2x} - 1}\right) + C$

$= -\frac{\mathrm{e}^{2x} + 1}{\mathrm{e}^{2x} - 1} + C$

$= -\frac{\mathrm{e}^x + \mathrm{e}^{-x}}{\mathrm{e}^x - \mathrm{e}^{-x}} + C$

$= -\coth x + C$

7) a)

$\frac{\mathrm{d}y}{\mathrm{d}x} = \operatorname{cosech} 3x \coth 3x$

$\Rightarrow y = -\frac{1}{3}\operatorname{cosech} 3x + c$

7) b)

$\frac{\mathrm{d}y}{\mathrm{d}x} = 3\operatorname{cosech}^2 \frac{x}{9}$

$\Rightarrow y = 3 \times 9 \times -\coth \frac{x}{9} + c = -27 \coth \frac{x}{9} + c$

7) c)

$\frac{\mathrm{d}y}{\mathrm{d}x} = 6 \operatorname{sech} 9x \tanh 9x$

$\Rightarrow y = 6 \times \frac{1}{9} \times -\operatorname{sech} 9x + c = -\frac{2}{3}\operatorname{sech} 9x + c$

7) d)

$\frac{\mathrm{d}y}{\mathrm{d}x} = 10 \coth 2x$

$\Rightarrow y = 10 \times \frac{1}{2}\ln|\sinh 2x| + c = 5 \ln|\sinh 2x| + c$

8)

$\int \frac{\cosh x}{\sinh^2 x + 1}\mathrm{d}x$

Use Integration by Substitution:

Let $u = \sinh x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \cosh x \Rightarrow \mathrm{d}x = \frac{1}{\cosh x}\mathrm{d}u$

$\int \frac{\cosh x}{\sinh^2 x + 1}\mathrm{d}x = \int \frac{1}{u^2 + 1}\mathrm{d}u = \arctan u + c = \arctan(\sinh x) + c$

9)

$\int \frac{\cos x}{\sqrt{\sin^2 x + 1}}\mathrm{d}x$

Use Integration by Substitution:

Let $u = \sin x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \cos x \Rightarrow \mathrm{d}x = \frac{1}{\cos x}\mathrm{d}u$

$\int \frac{\cos x}{\sqrt{\sin^2 x + 1}}\mathrm{d}x = \int \frac{1}{\sqrt{u^2 + 1}}\mathrm{d}u = \operatorname{arsinh} u + c = \operatorname{arsinh}(\sin x) + c$

10) a)

$u = \cosh^2 x = (\cosh x)^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \sinh x \times 2(\cosh x)^1 = 2\sinh x \cosh x = \sinh 2x$

10) b)

$\int \frac{\sinh 2x}{\cosh^4 x+1}\,\mathrm{d}x$

Use Integration by Substitution:

Let $u = \cosh^2 x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \sinh 2x \Rightarrow \mathrm{d}x = \frac{1}{\sinh 2x}\,\mathrm{d}u$

$\therefore \int \frac{\sinh 2x}{\cosh^4 x+1}\,\mathrm{d}x = \int \frac{1}{u^2+1}\,\mathrm{d}u = \arctan u + c = \arctan(\cosh^2 x) + c$

11) a)

$\int \operatorname{arsinh} x\,\mathrm{d}x = \int 1 \times \operatorname{arsinh} x\,\mathrm{d}x$

Use Integration by Parts:

$u = \operatorname{arsinh} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{x^2+1}} \qquad v = x$

$\therefore \int \operatorname{arsinh} x\,\mathrm{d}x = x\operatorname{arsinh} x - \int \frac{x}{\sqrt{x^2+1}}\,\mathrm{d}x$

For $\int \frac{x}{\sqrt{x^2+1}}\,\mathrm{d}x$ use Integration by Substitution:

Let $u = x^2 + 1$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x$

$\Rightarrow \mathrm{d}x = \frac{1}{2x}\,\mathrm{d}u$

$\therefore \int \frac{x}{\sqrt{x^2+1}}\,\mathrm{d}x = \int \frac{x}{\sqrt{u}} \times \frac{1}{2x}\,\mathrm{d}u = \frac{1}{2}\int u^{-\frac{1}{2}}\,\mathrm{d}u = u^{\frac{1}{2}} + c = \sqrt{x^2+1} + c$

So:

$\int \operatorname{arsinh} x\,\mathrm{d}x = x\operatorname{arsinh} x - \sqrt{x^2+1} + c$

11) b)

$\int \operatorname{arcosh} x \, \mathrm{d}x = \int 1 \times \operatorname{arcosh} x \, \mathrm{d}x$

Use Integration by Parts:

$u = \operatorname{arcosh} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{x^2-1}} \qquad v = x$

$\therefore \int \operatorname{arcosh} x \, \mathrm{d}x = x \operatorname{arcosh} x - \int \frac{x}{\sqrt{x^2-1}} \mathrm{d}x$

For $\int \frac{x}{\sqrt{x^2-1}} \mathrm{d}x$ use Integration by Substitution:

Let $u = x^2 - 1$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x$

$\Rightarrow \mathrm{d}x = \frac{1}{2x} \mathrm{d}u$

$\therefore \int \frac{x}{\sqrt{x^2-1}} \mathrm{d}x = \int \frac{x}{\sqrt{u}} \times \frac{1}{2x} \mathrm{d}u = \frac{1}{2} \int u^{-\frac{1}{2}} \mathrm{d}u = u^{\frac{1}{2}} + c = \sqrt{x^2-1} + c$

So:

$\int \operatorname{arcosh} x \, \mathrm{d}x = x \operatorname{arcosh} x - \sqrt{x^2-1} + c$

11) c)

$\int \operatorname{artanh} x \, \mathrm{d}x = \int 1 \times \operatorname{artanh} x \, \mathrm{d}x$

Use Integration by Parts:

$u = \operatorname{artanh} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1-x^2} \qquad v = x$

$\therefore \int \operatorname{artanh} x \, \mathrm{d}x = x \operatorname{artanh} x - \int \frac{x}{1-x^2} \mathrm{d}x$

For $\int \frac{x}{1-x^2} \mathrm{d}x$ use Integration by Substitution (or use $\int \frac{\mathrm{f}'(x)}{\mathrm{f}(x)} \mathrm{d}x = |\mathrm{f}(x)| + c$):

Let $u = 1 - x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -2x$

$\Rightarrow \mathrm{d}x = -\frac{1}{2x} \mathrm{d}u$

$\therefore \int \frac{x}{1-x^2} \mathrm{d}x = \int \frac{x}{u} \times -\frac{1}{2x} \mathrm{d}u = -\frac{1}{2} \int \frac{1}{u} \mathrm{d}u = -\frac{1}{2} \ln|u| + c = -\frac{1}{2} \ln|1-x^2| + c$

So:

$\int \operatorname{artanh} x \, \mathrm{d}x = x \operatorname{artanh} x + \frac{1}{2} \ln|1-x^2| + c$

12)

$\int_0^{\frac{1}{3}} 6 \operatorname{artanh} x \, \mathrm{d}x$

Use Integration by Parts:

$u = \operatorname{artanh} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 6$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1-x^2} \qquad v = 6x$

$\therefore \int 6 \operatorname{artanh} x \, \mathrm{d}x = 6x \operatorname{artanh} x - \int \frac{6x}{1-x^2} \mathrm{d}x$

For $\int \frac{6x}{1-x^2} \mathrm{d}x$ use Integration by Substitution (or use $\int \frac{\mathrm{f}'(x)}{\mathrm{f}(x)} \mathrm{d}x = |\mathrm{f}(x)| + c$):

Let $u = 1 - x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -2x$

$\Rightarrow \mathrm{d}x = -\frac{1}{2x} \mathrm{d}u$

$\therefore \int \frac{6x}{1-x^2} \mathrm{d}x = \int \frac{6x}{u} \times -\frac{1}{2x} \mathrm{d}u = -3 \int \frac{1}{u} \mathrm{d}u = -3 \ln|u| + c = -3 \ln|1 - x^2| + c$

So:

$\int_0^{\frac{1}{3}} 6 \operatorname{artanh} x \, \mathrm{d}x = [6x \operatorname{artanh} x + 3 \ln|1 - x^2|]_0^{\frac{1}{3}}$

$= \left[6x \times \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) + 3 \ln|1 - x^2|\right]_0^{\frac{1}{3}}$

$= \left[3x \ln\left(\frac{1+x}{1-x}\right) + 3 \ln|1 - x^2|\right]_0^{\frac{1}{3}}$

$= \left(3 \times \frac{1}{3} \times \ln\left(\frac{1+\frac{1}{3}}{1-\frac{1}{3}}\right) + 3 \ln\left|1 - \left(\frac{1}{3}\right)^2\right|\right) - \left(3 \times 0 \times \ln\left(\frac{1+0}{1-0}\right) + 3 \ln|1 - 0|\right)$

$= \left(\ln 2 + 3 \ln \frac{8}{9}\right) - 0$

$= \ln 2 + 3 \ln\left(\frac{2^3}{3^2}\right)$

$= \ln 2 + \ln\left(\frac{2^9}{3^6}\right)$

$= \ln\left(2 \times \frac{2^9}{3^6}\right)$

$= \ln\left(\frac{2^{10}}{3^6}\right)$

13)

$\int_0^{\frac{\tanh 4}{2}} 4 \operatorname{artanh} 2x \, \mathrm{d}x$

Use Integration by Parts:

$u = \operatorname{artanh} 2x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 4$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2}{1-4x^2} \qquad v = 4x$

$\therefore \int 4 \operatorname{artanh} 2x \, \mathrm{d}x = 4x \operatorname{artanh} 2x - \int \frac{8x}{1-4x^2} \mathrm{d}x$

For $\int \frac{8x}{1-4x^2} \mathrm{d}x$ use Integration by Substitution (or use $\int \frac{\mathrm{f}'(x)}{\mathrm{f}(x)} \mathrm{d}x = |\mathrm{f}(x)| + c$:

Let $u = 1 - 4x^2$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -8x$

$\Rightarrow \mathrm{d}x = -\frac{1}{8x} \mathrm{d}u$

$\therefore \int \frac{8x}{1-4x^2} \mathrm{d}x = \int \frac{8x}{u} \times -\frac{1}{8x} \mathrm{d}u = -\int \frac{1}{u} \mathrm{d}u = -\ln|u| + c = -\ln|1-4x^2| + c$

So:

$\int_0^{\frac{\tanh 4}{2}} 4 \operatorname{artanh} 2x \, \mathrm{d}x = [4x \operatorname{artanh} 2x + \ln|1-4x^2|]_0^{\frac{\tanh 4}{2}}$

$= \left[4x \times \frac{1}{2} \ln\left(\frac{1+2x}{1-2x}\right) + \ln|1-4x^2|\right]_0^{\frac{\tanh 4}{2}}$

$= \left[2x \ln\left(\frac{1+2x}{1-2x}\right) + \ln|1-4x^2|\right]_0^{\frac{\tanh 4}{2}}$

$= \left(2 \times \frac{\tanh 4}{2} \times \ln\left(\frac{1+2\times\frac{\tanh 4}{2}}{1-2\times\frac{\tanh 4}{2}}\right) + \ln\left|1 - 4\left(\frac{\tanh 4}{2}\right)^2\right|\right) - \left(2 \times 0 \times \ln\left(\frac{1+2\times 0}{1-2\times 0}\right) + \ln|1 - 4 \times 0^2|\right)$

$= \left(\tanh 4 \times \ln\left(\frac{1+\tanh 4}{1-\tanh 4}\right) + \ln|1-\tanh^2 4|\right) - 0$

$= \tanh 4 \times \ln\left(\frac{1+\frac{e^4-e^{-4}}{e^4+e^{-4}}}{1-\frac{e^4-e^{-4}}{e^4+e^{-4}}}\right) + \ln|1-\tanh^2 4|$

$= \tanh 4 \times \ln\left(\frac{1+\frac{e^8-1}{e^8+1}}{1-\frac{e^8-1}{e^8+1}}\right) + \ln|1-\tanh^2 4|$

$= \tanh 4 \times \ln\left(\frac{e^8+1+e^8-1}{e^8+1-(e^8-1)}\right) + \ln|1-\tanh^2 4|$

$= \tanh 4 \times \ln\left(\frac{2e^8}{2}\right) + \ln|1-\tanh^2 4|$

$= \tanh 4 \times \ln e^8 + \ln|1-\tanh^2 4|$

$= 8 \tanh 4 + \ln|1-\tanh^2 4|$

14) a)

$\frac{\mathrm{d}}{\mathrm{d}x}(x \operatorname{arcosh} 3x) = x \times 3 \times \frac{1}{\sqrt{(3x)^2-1}} + \operatorname{arcosh} 3x$

$= \frac{3x}{\sqrt{9x^2-1}} + \operatorname{arcosh} 3x$

14) b)

From part a.:

$\int \left(\frac{3x}{\sqrt{9x^2-1}} + \operatorname{arcosh} 3x\right) \mathrm{d}x = x \operatorname{arcosh} 3x + c$

$\Rightarrow \int \frac{3x}{\sqrt{9x^2-1}} \mathrm{d}x + \int \operatorname{arcosh} 3x \, \mathrm{d}x = x \operatorname{arcosh} 3x + c$

$\Rightarrow \int \operatorname{arcosh} 3x \, \mathrm{d}x = x \operatorname{arcosh} 3x - \int \frac{3x}{\sqrt{9x^2-1}} \mathrm{d}x + c$

Use Integration by Substitution:

Let $u = 9x^2 - 1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 18x \Rightarrow \mathrm{d}x = \frac{1}{18x}\mathrm{d}u$

$\therefore \int \frac{3x}{\sqrt{9x^2-1}} \mathrm{d}x = \int \frac{3x}{\sqrt{u}} \times \frac{1}{18x} \mathrm{d}u = \frac{1}{6}\int \frac{1}{\sqrt{u}} \mathrm{d}u = \frac{1}{6}\int u^{-\frac{1}{2}} \mathrm{d}u = \frac{1}{6} \times \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + c = \frac{1}{3}u^{\frac{1}{2}} + c$

$= \frac{1}{3}\sqrt{9x^2-1} + c$

So:

$\int \operatorname{arcosh} 3x \, \mathrm{d}x = x \operatorname{arcosh} 3x - \frac{1}{3}\sqrt{9x^2-1} + c$

15)

$\int x^2 \operatorname{arcosh} x \, \mathrm{d}x$

Use Integration by Parts:

$u = \operatorname{arcosh} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x^2$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{x^2-1}} \qquad v = \frac{1}{3}x^3$

$\therefore \int x^2 \operatorname{arcosh} x \, \mathrm{d}x = \frac{1}{3}x^3 \operatorname{arcosh} x - \frac{1}{3}\int \frac{x^3}{\sqrt{x^2-1}} \mathrm{d}x$

For $\int \frac{x^3}{\sqrt{x^2-1}} \mathrm{d}x$ use Integration by Substitution:

Let $u = x^2 - 1 \Rightarrow x^2 = u + 1$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x$

$\Rightarrow \mathrm{d}x = \frac{1}{2x}\mathrm{d}u$

$\therefore \int \frac{x^3}{\sqrt{x^2-1}} \mathrm{d}x = \int \frac{x^3}{\sqrt{u}} \times \frac{1}{2x} \mathrm{d}u = \frac{1}{2}\int x^2 u^{-\frac{1}{2}} \mathrm{d}u = \frac{1}{2}\int (u+1)u^{-\frac{1}{2}} \mathrm{d}u = \frac{1}{2}\int \left(u^{\frac{1}{2}} + u^{-\frac{1}{2}}\right) \mathrm{d}u =$

$\frac{1}{2}\left(\frac{2}{3}u^{\frac{3}{2}} + 2u^{\frac{1}{2}}\right) + c = \frac{1}{3}u^{\frac{3}{2}} + u^{\frac{1}{2}} + c = \frac{1}{3}(x^2-1)^{\frac{3}{2}} + \sqrt{x^2-1} + c$

So:

$\int x^2 \operatorname{arcosh} x \, \mathrm{d}x = \frac{1}{3}x^3 \operatorname{arcosh} x - \frac{1}{3}\left(\frac{1}{3}(x^2-1)^{\frac{3}{2}} + \sqrt{x^2-1}\right) + c$

$= \frac{1}{3}x^3 \operatorname{arcosh} x - \frac{1}{9}(x^2-1)^{\frac{3}{2}} - \frac{1}{3}\sqrt{x^2-1} + c$

16) a)

$\frac{2x^2}{1-x^2} \equiv \frac{2x^2-2+2}{1-x^2} \equiv \frac{2x^2-2}{1-x^2} + \frac{2}{1-x^2} \equiv \frac{-2(1-x^2)}{1-x^2} + \frac{2}{1-x^2} \equiv -2 + \frac{2}{1-x^2}$

Or using polynomial division:

	-2	
$-x^2$	$2x^2$	$0x$
$+0x$	$0x$	2
$+1$	-2	

$\therefore \frac{2x^2}{1-x^2} \equiv -2 + \frac{2}{1-x^2}$

$\therefore A = -2$ and $B = 2$

16) b)

$\int 4x \operatorname{artanh} x \, \mathrm{d}x$

Use Integration by Parts:

$u = \operatorname{artanh} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = 4x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1-x^2} \qquad v = 2x^2$

$\therefore \int 4x \operatorname{artanh} x \, \mathrm{d}x = 2x^2 \operatorname{artanh} x - \int \frac{2x^2}{1-x^2} \mathrm{d}x$

Consider $\int \frac{2x^2}{1-x^2} \mathrm{d}x = \int \left(-2 + \frac{2}{1-x^2}\right) \mathrm{d}x$

$= -2x + 2 \operatorname{artanh} x + c$

So:

$\int 4x \operatorname{artanh} x \, \mathrm{d}x = 2x^2 \operatorname{artanh} x - (-2x + 2 \operatorname{artanh} x) + c$

$= 2x^2 \operatorname{artanh} x + 2x - 2 \operatorname{artanh} x + c$

17) a)

$\int \frac{\operatorname{arsinh} x}{\sqrt{x^2+1}} \mathrm{d}x$

Use Integration by Parts:

$u = \operatorname{arsinh} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1}{\sqrt{x^2+1}}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{x^2+1}} \qquad v = \operatorname{arsinh} x$

$\therefore \int \frac{\operatorname{arsinh} x}{\sqrt{x^2+1}} \mathrm{d}x = (\operatorname{arsinh} x)^2 - \int \frac{\operatorname{arsinh} x}{\sqrt{x^2+1}} \mathrm{d}x$

$\Rightarrow 2 \int \frac{\operatorname{arsinh} x}{\sqrt{x^2+1}} \mathrm{d}x = (\operatorname{arsinh} x)^2$

$\therefore \int \frac{\operatorname{arsinh} x}{\sqrt{x^2+1}} \mathrm{d}x = \frac{1}{2}(\operatorname{arsinh} x)^2 + c$

17) b)

$\int \frac{\operatorname{arcosh} x}{\sqrt{x^2-1}}\,\mathrm{d}x$

Use Integration by Parts:

$u = \operatorname{arcosh} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1}{\sqrt{x^2-1}}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{x^2-1}} \qquad v = \operatorname{arcosh} x$

$\therefore \int \frac{\operatorname{arcosh} x}{\sqrt{x^2-1}}\,\mathrm{d}x = (\operatorname{arcosh} x)^2 - \int \frac{\operatorname{arcosh} x}{\sqrt{x^2-1}}\,\mathrm{d}x$

$\Rightarrow 2\int \frac{\operatorname{arcosh} x}{\sqrt{x^2-1}}\,\mathrm{d}x = (\operatorname{arcosh} x)^2$

$\therefore \int \frac{\operatorname{arcosh} x}{\sqrt{x^2-1}}\,\mathrm{d}x = \frac{1}{2}(\operatorname{arcosh} x)^2 + c$

17) c)

$\int \frac{\operatorname{artanh} x}{1-x^2}\,\mathrm{d}x$

Use Integration by Parts:

$u = \operatorname{artanh} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1}{1-x^2}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1-x^2} \qquad v = \operatorname{artanh} x$

$\therefore \int \frac{\operatorname{artanh} x}{1-x^2}\,\mathrm{d}x = (\operatorname{artanh} x)^2 - \int \frac{\operatorname{artanh} x}{1-x^2}\,\mathrm{d}x$

$\Rightarrow 2\int \frac{\operatorname{artanh} x}{1-x^2}\,\mathrm{d}x = (\operatorname{artanh} x)^2$

$\therefore \int \frac{\operatorname{artanh} x}{1-x^2}\,\mathrm{d}x = \frac{1}{2}(\operatorname{artanh} x)^2 + c$

18)

$\int_0^2 \frac{\operatorname{arsinh} 2x}{\sqrt{4x^2+1}}\,\mathrm{d}x$

Use Integration by Parts:

$$u = \operatorname{arsinh} 2x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1}{\sqrt{4x^2+1}}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2}{\sqrt{4x^2+1}} \qquad v = \frac{1}{2}\operatorname{arsinh} 2x$$

$$\int \frac{\operatorname{arsinh} 2x}{\sqrt{4x^2+1}}\,\mathrm{d}x = \frac{1}{2}(\operatorname{arsinh} 2x)^2 - \int \frac{\operatorname{arsinh} 2x}{\sqrt{4x^2+1}}\,\mathrm{d}x$$

$$\Rightarrow 2\int \frac{\operatorname{arsinh} 2x}{\sqrt{4x^2+1}}\,\mathrm{d}x = \frac{1}{2}(\operatorname{arsinh} 2x)^2$$

$$\therefore \int \frac{\operatorname{arsinh} 2x}{\sqrt{4x^2+1}}\,\mathrm{d}x = \frac{1}{4}(\operatorname{arsinh} 2x)^2 + c$$

$$\therefore \int_0^2 \frac{\operatorname{arsinh} 2x}{\sqrt{4x^2+1}}\,\mathrm{d}x = \left[\frac{1}{4}(\operatorname{arsinh} 2x)^2\right]_0^2$$

$$= \frac{1}{4}(\operatorname{arsinh} 4)^2 - \frac{1}{4}(\operatorname{arsinh} 0)^2$$

$$= \frac{1}{4}\left(\ln\left(4+\sqrt{4^2+1}\right)\right)^2 - 0$$

$$= \frac{1}{4}\left(\ln\left(4+\sqrt{17}\right)\right)^2$$

19)

$\int_{\frac{1}{3}}^4 \frac{\operatorname{arcosh} 3x}{\sqrt{9x^2-1}}\,\mathrm{d}x$

Use Integration by Parts:

$$u = \operatorname{arcosh} 3x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1}{\sqrt{9x^2-1}}$$

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{3}{\sqrt{9x^2-1}} \qquad v = \frac{1}{3}\operatorname{arcosh} 3x$$

$$\int \frac{\operatorname{arcosh} 3x}{\sqrt{9x^2-1}}\,\mathrm{d}x = \frac{1}{3}(\operatorname{arcosh} 3x)^2 - \int \frac{\operatorname{arcosh} 3x}{\sqrt{9x^2-1}}\,\mathrm{d}x$$

$$\Rightarrow 2\int \frac{\operatorname{arcosh} 3x}{\sqrt{9x^2-1}}\,\mathrm{d}x = \frac{1}{3}(\operatorname{arcosh} 3x)^2$$

$$\therefore \int \frac{\operatorname{arcosh} 3x}{\sqrt{9x^2-1}}\,\mathrm{d}x = \frac{1}{6}(\operatorname{arcosh} 3x)^2 + c$$

$$\therefore \int_{\frac{1}{3}}^4 \frac{\operatorname{arcosh} 3x}{\sqrt{9x^2-1}}\,\mathrm{d}x = \left[\frac{1}{6}(\operatorname{arcosh} 3x)^2\right]_{\frac{1}{3}}^4$$

$$= \frac{1}{6}(\operatorname{arcosh} 12)^2 - \frac{1}{6}(\operatorname{arcosh} 1)^2$$

$$= \frac{1}{6}\left(\ln\left(12+\sqrt{12^2-1}\right)\right)^2 - 0$$

$$= \frac{1}{6}\left(\ln\left(12+\sqrt{143}\right)\right)^2$$

20) a)

$$\frac{10}{5\cosh x - 4\sinh x}$$
$$= \frac{10}{5\times\frac{e^x+e^{-x}}{2} - 4\times\frac{e^x-e^{-x}}{2}}$$
$$= \frac{10}{\frac{5}{2}(e^x+e^{-x}) - 2(e^x-e^{-x})}$$
$$= \frac{10}{\frac{1}{2}e^x + \frac{9}{2}e^{-x}}$$
$$= \frac{20}{e^x+9e^{-x}}$$
$$= \frac{20e^x}{e^{2x}+9}$$

20) b)

$$\int_0^{\ln 3} \frac{30}{5\cosh x - 4\sinh x}\,dx = \int_0^{\ln 3} \frac{60e^x}{e^{2x}+9}\,dx$$

Use Integration by Substitution:

Let $u = e^x \Rightarrow \frac{du}{dx} = e^x \Rightarrow dx = \frac{1}{e^x}du$

Change limits:

$x = \ln 3 \Rightarrow u = e^{\ln 3} = 3$

$x = 0 \Rightarrow u = e^0 = 1$

$$\therefore \int_0^{\ln 3} \frac{60e^x}{e^{2x}+9}\,dx$$
$$= \int_1^3 \frac{60}{u^2+9}\,du$$
$$= \left[20\arctan\left(\frac{1}{3}x\right)\right]_1^3$$
$$= (20\arctan 1) - \left(20\arctan\frac{1}{3}\right)$$
$$= \left(20\times\frac{\pi}{4}\right) - \left(20\times\arctan\frac{1}{3}\right)$$
$$= 5\pi - 20\arctan\frac{1}{3}$$

21)

$$\int \frac{1}{(1-x^2)((\operatorname{artanh} x)^2+1)}\,dx$$

Use Integration by Substitution:

$u = \operatorname{artanh} x \Rightarrow \frac{du}{dx} = \frac{1}{1-x^2} \Rightarrow dx = (1-x^2)du$

$$\therefore \int \frac{1}{(1-x^2)((\operatorname{artanh} x)^2+1)}\,dx = \int \frac{1}{u^2+1}\,du = \arctan u + c$$
$$= \arctan(\operatorname{artanh} x) + c$$

22) a)

$y = \operatorname{arcosh}\left(\frac{\sqrt{x}}{4}\right) = \operatorname{arcosh}\left(\frac{1}{4}x^{\frac{1}{2}}\right)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{8}x^{-\frac{1}{2}} \times \frac{1}{\sqrt{\left(\frac{1}{4}x^{\frac{1}{2}}\right)^2 - 1}}$

$= \frac{1}{8\sqrt{x}} \times \frac{1}{\sqrt{\frac{1}{16}x - 1}}$

$= \frac{1}{8\sqrt{x}} \times \frac{1}{\sqrt{\frac{1}{16}(x-16)}}$

$= \frac{1}{8\sqrt{x}} \times \frac{1}{\frac{1}{4}\sqrt{x-16}}$

$= \frac{1}{2\sqrt{x(x-16)}}$

22) b)

$\int_{64}^{144} \frac{1}{\sqrt{x(x-16)}}\,\mathrm{d}x$

$= \left[2\operatorname{arcosh}\left(\frac{\sqrt{x}}{4}\right)\right]_{64}^{144}$

$= (2\operatorname{arcosh} 3) - (2\operatorname{arcosh} 2)$

$= 2(\operatorname{arcosh} 3 - \operatorname{arcosh} 2)$

$= 2\left(\ln\left(3 + \sqrt{3^2 - 1}\right) - \ln\left(2 + \sqrt{2^2 - 1}\right)\right)$

$= 2\left(\ln\left(3 + 2\sqrt{2}\right) - \ln\left(2 + \sqrt{3}\right)\right)$

$= 2\ln\left(\frac{3+2\sqrt{2}}{2+\sqrt{3}}\right)$

23)

$\int \frac{x \operatorname{arsinh} x}{\sqrt{x^2+1}}\,\mathrm{d}x$

Use Integration by Parts:

$u = \operatorname{arsinh} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \frac{x}{\sqrt{x^2+1}}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{x^2+1}} \qquad v = \sqrt{x^2+1}$

$\therefore \int \frac{x \operatorname{arsinh} x}{\sqrt{x^2+1}}\,\mathrm{d}x = \sqrt{x^2+1}\operatorname{arsinh} x - \int 1\,\mathrm{d}x$

$= \sqrt{x^2+1}\operatorname{arsinh} x - x + c$

24) a)

$\frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{arcosh} 2x) = 2 \times \frac{1}{\sqrt{(2x)^2-1}} = \frac{2}{\sqrt{4x^2-1}}$

24) b)

$\int \frac{x \operatorname{arcosh} 2x}{\sqrt{4x^2-1}} \mathrm{d}x$

Use Integration by Parts:

$u = \operatorname{arcosh} 2x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \frac{x}{\sqrt{4x^2-1}}$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2}{\sqrt{4x^2-1}} \qquad v = \frac{1}{4}\sqrt{4x^2-1}$

$\therefore \int \frac{x \operatorname{arcosh} 2x}{\sqrt{4x^2-1}} \mathrm{d}x = \frac{1}{4}\sqrt{4x^2-1} \operatorname{arcosh} 2x - \int \frac{1}{2} \mathrm{d}x$

$= \frac{1}{4}\sqrt{4x^2-1} \operatorname{arcosh} 2x - \frac{1}{2}x + c$

25) a)

Use $\sinh 2A = 2 \sinh A \cosh A$

$= 2 \sinh A \sqrt{\sinh^2 A + 1}$

$\therefore \sinh(2 \operatorname{arsinh} x) = 2 \sinh(\operatorname{arsinh} x) \sqrt{\sinh^2(\operatorname{arsinh} x) + 1}$

$= 2x\sqrt{x^2+1}$ as required

Alternatively:

$\sinh(2 \operatorname{arsinh} x)$

$= 2 \sinh(\operatorname{arsinh} x) \cosh(\operatorname{arsinh} x)$

$= 2x \cosh(\operatorname{arsinh} x)$

But $y = \operatorname{arsinh} x \Rightarrow \sinh y = x \Rightarrow 1 + \sinh^2 y = 1 + x^2 \Rightarrow \cosh y = \sqrt{1+x^2}$

$\Rightarrow \cosh(\operatorname{arsinh} x) = \sqrt{1+x^2}$

$\therefore 2x\sqrt{x^2+1}$ as req.

25) b)

$\int x \operatorname{arsinh} x \, \mathrm{d}x$

Use Integration by Parts:

$u = \operatorname{arsinh} x \qquad \frac{\mathrm{d}v}{\mathrm{d}x} = x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{\sqrt{x^2+1}} \qquad v = \frac{1}{2}x^2$

$\therefore \int x \operatorname{arsinh} x \, \mathrm{d}x = \frac{1}{2}x^2 \operatorname{arsinh} x - \frac{1}{2}\int \frac{x^2}{\sqrt{x^2+1}} \mathrm{d}x$

For $\int \frac{x^2}{\sqrt{x^2+1}} \mathrm{d}x$ use Integration by Substitution:

Let $x = \sinh u$

$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cosh u$

$\Rightarrow \mathrm{d}x = \cosh u \, \mathrm{d}u$

$\therefore \int \frac{x^2}{\sqrt{x^2+1}} \mathrm{d}x = \int \frac{\sinh^2 u}{\sqrt{\sinh^2 u+1}} \times \cosh u \, \mathrm{d}u = \int \frac{\sinh^2 u \cosh u}{\sqrt{\cosh^2 u}} \mathrm{d}u = \int \frac{\sinh^2 u \cosh u}{\cosh u} \mathrm{d}u$

$= \int \sinh^2 u \, \mathrm{d}u = \int \frac{1}{2}(\cosh 2u - 1) \, \mathrm{d}u = \frac{1}{4}\sinh 2u - \frac{1}{2}u + c$

$= \frac{1}{4}\sinh(2 \operatorname{arsinh} x) - \frac{1}{2}\operatorname{arsinh} x + c$

$= \frac{1}{4} \times 2x\sqrt{x^2+1} - \frac{1}{2}\operatorname{arsinh} x + c$

$= \frac{1}{2}x\sqrt{x^2+1} - \frac{1}{2}\operatorname{arsinh} x + c$

So:

$\int x \operatorname{arsinh} x \, \mathrm{d}x = \frac{1}{2}x^2 \operatorname{arsinh} x - \frac{1}{2}\left(\frac{1}{2}x\sqrt{x^2+1} - \frac{1}{2}\operatorname{arsinh} x\right) + c$

$= \frac{1}{2}x^2 \operatorname{arsinh} x - \frac{1}{4}x\sqrt{x^2+1} + \frac{1}{4}\operatorname{arsinh} x + c$

26) a)

$\int \operatorname{cosech} x \, \mathrm{d}x$

$= \int \frac{1}{\sinh x} \mathrm{d}x$

$= \int \frac{1}{\frac{\mathrm{e}^x - \mathrm{e}^{-x}}{2}} \mathrm{d}x$

$= \int \frac{2}{\mathrm{e}^x - \mathrm{e}^{-x}} \mathrm{d}x$

$= \int \frac{2\mathrm{e}^x}{\mathrm{e}^{2x} - 1} \mathrm{d}x$

Use Integration by Substitution:

Let $u = \mathrm{e}^x$ then $\frac{\mathrm{d}u}{\mathrm{d}x} = \mathrm{e}^x \Rightarrow \mathrm{d}x = \frac{1}{\mathrm{e}^x} \mathrm{d}u$

$\therefore \int \frac{2\mathrm{e}^x}{\mathrm{e}^{2x} - 1} \mathrm{d}x = \int \frac{2}{u^2 - 1} \mathrm{d}u$

Use Partial Fractions:

$\frac{2}{u^2 - 1} \equiv \frac{2}{(u-1)(u+1)} \equiv \frac{A}{u-1} + \frac{B}{u+1}$

$2 \equiv A(u+1) + B(u-1)$

Let $u = 1 \Rightarrow 2 = 2A \Rightarrow A = 1$

Let $u = -1 \Rightarrow 2 = -2B \Rightarrow B = -1$

$\therefore \int \frac{2}{u^2 - 1} \mathrm{d}u = \int \left(\frac{1}{u-1} - \frac{1}{u+1} \right) \mathrm{d}u$

$= \ln|u - 1| - \ln|u + 1| + c$

$= \ln \left| \frac{u-1}{u+1} \right| + c$

$= \ln \left| \frac{\mathrm{e}^x - 1}{\mathrm{e}^x + 1} \right| + c$

$= \ln \left| \tanh \frac{x}{2} \right| + c$

26) b) i)

$\int \operatorname{sech} x \, \mathrm{d}x$

$= \int \frac{1}{\cosh x} \mathrm{d}x$

$= \int \frac{1}{\frac{\mathrm{e}^x + \mathrm{e}^{-x}}{2}} \mathrm{d}x$

$= \int \frac{2}{\mathrm{e}^x + \mathrm{e}^{-x}} \mathrm{d}x$

$= \int \frac{2\mathrm{e}^x}{\mathrm{e}^{2x}+1} \mathrm{d}x$

Use Integration by Substitution:

Let $u = \mathrm{e}^x$ then $\frac{\mathrm{d}u}{\mathrm{d}x} = \mathrm{e}^x \Rightarrow \mathrm{d}x = \frac{1}{\mathrm{e}^x}\mathrm{d}u$

$\therefore \int \frac{2\mathrm{e}^x}{\mathrm{e}^{2x}+1} \mathrm{d}x = \int \frac{2}{u^2+1} \mathrm{d}u$

$= 2 \arctan u + c$

$= 2 \arctan(\mathrm{e}^x) + c$

26) b) ii)

$1 + \tanh\frac{x}{2} = 1 + \frac{\mathrm{e}^{\frac{x}{2}} - \mathrm{e}^{-\frac{x}{2}}}{\mathrm{e}^{\frac{x}{2}} + \mathrm{e}^{-\frac{x}{2}}} = \frac{\mathrm{e}^{\frac{x}{2}} + \mathrm{e}^{-\frac{x}{2}}}{\mathrm{e}^{\frac{x}{2}} + \mathrm{e}^{-\frac{x}{2}}} + \frac{\mathrm{e}^{\frac{x}{2}} - \mathrm{e}^{-\frac{x}{2}}}{\mathrm{e}^{\frac{x}{2}} + \mathrm{e}^{-\frac{x}{2}}} = \frac{2\mathrm{e}^{\frac{x}{2}}}{\mathrm{e}^{\frac{x}{2}} + \mathrm{e}^{-\frac{x}{2}}}$

$1 - \tanh\frac{x}{2} = 1 - \frac{\mathrm{e}^{\frac{x}{2}} - \mathrm{e}^{-\frac{x}{2}}}{\mathrm{e}^{\frac{x}{2}} + \mathrm{e}^{-\frac{x}{2}}} = \frac{\mathrm{e}^{\frac{x}{2}} + \mathrm{e}^{-\frac{x}{2}}}{\mathrm{e}^{\frac{x}{2}} + \mathrm{e}^{-\frac{x}{2}}} - \frac{\mathrm{e}^{\frac{x}{2}} - \mathrm{e}^{-\frac{x}{2}}}{\mathrm{e}^{\frac{x}{2}} + \mathrm{e}^{-\frac{x}{2}}} = \frac{2\mathrm{e}^{-\frac{x}{2}}}{\mathrm{e}^{\frac{x}{2}} + \mathrm{e}^{-\frac{x}{2}}}$

$\frac{1+\tanh\frac{x}{2}}{1-\tanh\frac{x}{2}} = \frac{\frac{2\mathrm{e}^{\frac{x}{2}}}{\mathrm{e}^{\frac{x}{2}} + \mathrm{e}^{-\frac{x}{2}}}}{\frac{2\mathrm{e}^{-\frac{x}{2}}}{\mathrm{e}^{\frac{x}{2}} + \mathrm{e}^{-\frac{x}{2}}}} = \frac{2\mathrm{e}^{\frac{x}{2}}}{2\mathrm{e}^{-\frac{x}{2}}} = \frac{\mathrm{e}^{\frac{x}{2}}}{\mathrm{e}^{-\frac{x}{2}}} = \mathrm{e}^x$ as req.

Alternatively:

$\tanh\frac{x}{2} = \frac{\mathrm{e}^{\frac{x}{2}} - \mathrm{e}^{-\frac{x}{2}}}{\mathrm{e}^{\frac{x}{2}} + \mathrm{e}^{-\frac{x}{2}}} = \frac{\mathrm{e}^x - 1}{\mathrm{e}^x + 1}$

Solve for e^x:

$\mathrm{e}^x \tanh\frac{x}{2} + \tanh\frac{x}{2} = \mathrm{e}^x - 1$

$\Rightarrow \tanh\frac{x}{2} + 1 = \mathrm{e}^x - \mathrm{e}^x \tanh\frac{x}{2}$

$\Rightarrow \tanh\frac{x}{2} + 1 = \mathrm{e}^x \left(1 - \tanh\frac{x}{2}\right)$

$\Rightarrow \mathrm{e}^x = \frac{1+\tanh\frac{x}{2}}{1-\tanh\frac{x}{2}}$ as req.

26) b) iii)

By using $\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}$, with $A=\frac{\pi}{4}$ and $B=\theta$ we get:

$\tan\left(\frac{\pi}{4}+\theta\right)=\frac{\tan\frac{\pi}{4}+\tan\theta}{1-\tan\frac{\pi}{4}\tan\theta}=\frac{1+\tan\theta}{1-\tan\theta}$

So that, $\tan\left(\frac{\pi}{4}+\arctan\left(\tanh\frac{x}{2}\right)\right)=\frac{1+\tan\left(\arctan\left(\tanh\frac{x}{2}\right)\right)}{1-\tan\left(\arctan\left(\tanh\frac{x}{2}\right)\right)}=\frac{1+\tanh\frac{x}{2}}{1-\tanh\frac{x}{2}}=\mathrm{e}^{x}$

So we have $\tan\left(\frac{\pi}{4}+\arctan\left(\tanh\frac{x}{2}\right)\right)=\mathrm{e}^{x}$

$\Rightarrow\frac{\pi}{4}+\arctan\left(\tanh\frac{x}{2}\right)=\arctan(\mathrm{e}^{x})$

$\Rightarrow\frac{\pi}{2}+2\arctan\left(\tanh\frac{x}{2}\right)=2\arctan(\mathrm{e}^{x})$

$\therefore\int\operatorname{sech}x\,\mathrm{d}x=2\arctan(\mathrm{e}^{x})+c$

$=2\arctan\left(\tanh\frac{x}{2}\right)+\frac{\pi}{2}+c$

$=2\arctan\left(\tanh\frac{x}{2}\right)+C$

As $\frac{\pi}{2}$ is just a constant

27) a)

$\frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{sech}x)=\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{\cosh x}\right)$

$=\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{2}{\mathrm{e}^{x}+\mathrm{e}^{-x}}\right)$

$=\frac{(\mathrm{e}^{x}+\mathrm{e}^{-x})\times0-2\times(\mathrm{e}^{x}-\mathrm{e}^{-x})}{(\mathrm{e}^{x}+\mathrm{e}^{-x})^{2}}$

$=\frac{-2(\mathrm{e}^{x}-\mathrm{e}^{-x})}{(\mathrm{e}^{x}+\mathrm{e}^{-x})^{2}}$

$=-\frac{2}{\mathrm{e}^{x}+\mathrm{e}^{-x}}\times\frac{\mathrm{e}^{x}-\mathrm{e}^{-x}}{\mathrm{e}^{x}+\mathrm{e}^{-x}}$

$=-\frac{2}{\mathrm{e}^{x}+\mathrm{e}^{-x}}\times\frac{\frac{\mathrm{e}^{x}-\mathrm{e}^{-x}}{2}}{\frac{\mathrm{e}^{x}+\mathrm{e}^{-x}}{2}}$

$=-\frac{1}{\cosh x}\times\frac{\sinh x}{\cosh x}$

$=-\operatorname{sech}x\tanh x$

27) b)

$\int x \operatorname{sech} x \tanh x \, \mathrm{d}x$

Use Integration by Parts:

$u = x$ $\qquad \frac{\mathrm{d}v}{\mathrm{d}x} = \operatorname{sech} x \tanh x$

$\frac{\mathrm{d}u}{\mathrm{d}x} = 1$ $\qquad v = -\operatorname{sech} x$

$\therefore \int x \operatorname{sech} x \tanh x \, \mathrm{d}x = -x \operatorname{sech} x + \int \operatorname{sech} x \, \mathrm{d}x$

Now:

$\int \operatorname{sech} x \, \mathrm{d}x$

$= \int \frac{1}{\cosh x} \mathrm{d}x$

$= \int \frac{1}{\frac{\mathrm{e}^x + \mathrm{e}^{-x}}{2}} \mathrm{d}x$

$= \int \frac{2}{\mathrm{e}^x + \mathrm{e}^{-x}} \mathrm{d}x$

$= \int \frac{2\mathrm{e}^x}{\mathrm{e}^{2x} + 1} \mathrm{d}x$

Use Integration by Substitution:

Let $u = \mathrm{e}^x$ then $\frac{\mathrm{d}u}{\mathrm{d}x} = \mathrm{e}^x \Rightarrow \mathrm{d}x = \frac{1}{\mathrm{e}^x} \mathrm{d}u$

$\therefore \int \frac{2\mathrm{e}^x}{\mathrm{e}^{2x} + 1} \mathrm{d}x = \int \frac{2}{u^2 + 1} \mathrm{d}u$

$= 2 \arctan u + c$

$= 2 \arctan(\mathrm{e}^x) + c$

$\therefore \int x \operatorname{sech} x \tanh x \, \mathrm{d}x = -x \operatorname{sech} x + 2 \arctan(\mathrm{e}^x) + c$